

Supplementary Information for “Reliable comparison of severe felt-intensity classifiers with versioned and dependent event data”

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Complete pairwise model comparisons

Supplementary Table S1 reports every paired Brier-score comparison used to assess ranking stability. The main manuscript reports the comparisons that determine its conclusions: every learned baseline improves over no-skill, while B2–B4 and B2–B5 remain unresolved.

Protocol exclusions and deviations

Supplementary Table S2 lists all exclusions, unexecuted branches, and interpretation boundaries. These entries change no frozen result.

Table S1 Paired Brier-score comparisons among admissible baselines

Model A	Model B	A – B	95% CI	P (A better)	CI relation
B0	B1	0.0339	[0.0171, 0.0511]	0.000	excludes 0
B0	B2	0.0515	[0.0342, 0.0681]	0.000	excludes 0
B0	B4	0.0455	[0.0307, 0.0598]	0.000	excludes 0
B0	B5	0.0507	[0.0319, 0.0682]	0.000	excludes 0
B1	B2	0.0176	[0.0101, 0.0249]	0.000	excludes 0
B1	B4	0.0115	[0.0023, 0.0205]	0.009	excludes 0
B1	B5	0.0168	[0.0055, 0.0273]	0.001	excludes 0
B2	B4	-0.0060	[-0.0118, 0.0002]	0.970	includes 0
B2	B5	-0.0008	[-0.0078, 0.0068]	0.613	includes 0
B4	B5	0.0052	[0.0003, 0.0099]	0.017	excludes 0

Note. Lower Brier score is better, so a negative A – B difference favors model A. Intervals use 2,000 Monte Carlo bootstrap draws; near-zero endpoints are interpreted cautiously and not as stable ranking guarantees. Because no numerical equivalence margin was specified before analysis, intervals including zero are not labelled equivalent. B0 is the no-skill reference; B1 is magnitude-only logistic regression; B2 is source-only logistic regression; B4 is random forest; and B5 is XGBoost.

Temporal and geographic subgroup summaries

Supplementary Table S3 reports the complete temporal and geographic summaries of the 430 fixed temporal-holdout predictions. Geographic cells are descriptive subgroups, not leave-region-out evidence.

Table S2 Protocol exclusions, deviations, and interpretation boundaries

Decision	Treatment	Rationale
B3 expanded-metadata model	Omitted from temporal holdout	Includes event year, which reveals the temporal evaluation period.
Published DYFI intensity equation	Omitted as comparator	Needs unavailable site distance, tectonic-region inputs, and probability conversion.
Two prelisted quarantine IDs	No effect in version 1	Both are absent from the frozen snapshot; no released record is excluded.
Geographic transport evaluation	Not executed	Only temporal-holdout geographic subgroups are reported; no leave-region-out or generalization claim.
Aggregation-grid sensitivity	Not executed	Required but absent; only threshold sensitivity of fixed predictions is reported.
Uncertainty coverage	Partial deviation	The protocol requested intervals for every metric and the split contrast; only Brier-based intervals were retained.
Equivalence interpretation	Descriptive only	No numerical margin was specified; intervals are reported without equivalence labels.

Note. These decisions were recorded before, or disclosed with, the analysis. No result, subset, cutoff, or comparison was omitted because of its observed outcome.

Table S3 Temporal and geographic summaries within the temporal holdout

Subset	Events	Brier score	AUROC
Temporal 2023	250	0.185	0.797
Temporal 2024	180	0.174	0.777
Geographic cells within temporal holdout (82)	430	0.151 (med.)	–

Note. The geographic row partitions the same 430 temporal-holdout predictions into 82 occupied 10° cells. It is a descriptive subgroup summary, not a leave-region-out or external geographic evaluation. The entry is the median cell-level Brier score; cell values range from 0.002 to 0.825 and are included in the public release.